
Part 107 Study Guide

Complete Preparation Manual

The most comprehensive Part 107 study resource available. Every topic on the FAA Knowledge Test explained in detail with key facts highlighted for easy review. From regulations to weather, airspace to decision making - everything you need to pass.

20

Lessons

75

Sections

375

Key Facts

100%

Coverage

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Table of Contents

Chapter 1: Part 107 Core Regulations

- What is Part 107?
- Basic Operating Limitations
- Remote Pilot Certification
- Key Operational Rules
- Airspace and Authorization

Chapter 2: Weather Fundamentals

- Weather Elements and Flight Safety
- Wind and Its Effects
- Visibility and Atmospheric Conditions
- Clouds and Ceiling Altitude

Chapter 3: Density Altitude & Performance

- What is Density Altitude?
- Factors Affecting Density Altitude
- Performance Effects and Calculations

Chapter 4: Aeronautical Decision Making

- What is Aeronautical Decision Making?
- The DECIDE Model
- Risk Assessment and Mitigation

Chapter 5: Registration & Remote ID

- Drone Registration Requirements
- Remote ID Requirements
- Operator Registration and Tracking
- Liability and Insurance Considerations

Chapter 6: Reading METARs

- METAR Format and Elements
- Wind and Visibility Codes
- Weather Phenomena Codes
- Clouds, Temperature, and Altimeter

Chapter 7: Weight, Balance & Loading

- Weight and Balance Principles
- Calculating Center of Gravity
- Loading and Payload Considerations

Chapter 8: Crew Resource Management

- What is Crew Resource Management?
- Communication and Coordination
- Error Management and Challenge

Chapter 9: Waivers & Special Operations

- What is a Part 107 Waiver?
- Requesting a Waiver
- Common Waiver Categories
- Waiver Conditions and Compliance

Chapter 10: Reading TAFs

- TAF Format and Purpose
- TAF Interpretation and Change Groups

Using TAFs for Flight Planning
TAF Code Examples

Chapter 11: Physiology: Alcohol, Drugs & Fatigue

Alcohol and Flight Safety
Drugs and Medications
Fatigue and Alertness

Chapter 12: Night Operations & Ops Over People

Night Operations Rules
Night Flight Considerations
Operations Over People - Categories and Requirements
Over-People Operations Safety and Mitigation

Chapter 13: Thunderstorms & Hazardous Weather

Thunderstorm Formation and Stages
Hazards Within Thunderstorms
Identification and Avoidance
Other Hazardous Weather Phenomena

Chapter 14: Emergency Procedures

What Constitutes an Emergency?
Emergency Procedures and Recovery
Emergency Equipment and Resources

Chapter 15: Airspace Classification Overview

Seven Classes of Airspace
Controlled vs. Uncontrolled Airspace
Airspace Dimensions and Altitudes
Authorization and Communication Requirements

Chapter 16: Class B, C, D Airspace

Class B Airspace - Busy Airports
Class C Airspace - Medium Traffic Airports
Class D Airspace - Smaller Airports with Towers
Special Considerations for Class B, C, D

Chapter 17: Class E & G Airspace

Class E Airspace - General Controlled Airspace
Class G Airspace - Uncontrolled
Differences Between Controlled and Uncontrolled
Finding Your Airspace Class

Chapter 18: Special Use Airspace & TFRs

Special Use Airspace Overview
Military Training Routes (MTRs)
Military Operations Areas (MOAs)
Temporary Flight Restrictions (TFRs)

Chapter 19: Reading Sectional Charts

Chart Overview and Purpose
Airspace Depiction on Charts
Airports and Runways
Navigation Aids and Obstacles

Chapter 20: LAANC & Airspace Authorization

What is LAANC?
How LAANC Works

LAANC Coverage and Limitations

Pre-Flight Airspace Authorization Checklist

REGULATIONS

What is Part 107?

Part 107 is the FAA regulation that governs small Unmanned Aircraft System (sUAS) operations for commercial purposes. Established in 2016, Part 107 replaced the older Section 333 exemption process and established standardized rules for safe drone operations. Part 107 applies to drones weighing between 0.55 pounds (250 grams) and 55 pounds. The regulation is designed to enable commercial drone use while protecting people and property on the ground and maintaining airspace safety.

KEY FACTS

- Part 107 covers drones 0.55-55 pounds
- Established in 2016
- Applies to commercial operations
- Requires remote pilot certification

Basic Operating Limitations

Part 107 establishes clear operating limits that all remote pilots must follow. The altitude limit is 400 feet above ground level (AGL), measured from the highest point of ground within a 400-foot radius. The maximum horizontal speed is 100 mph. All operations must maintain visual line of sight (VLOS) with the drone, meaning the remote pilot or visual observer must be able to see the aircraft with unaided vision at all times. Visibility must be at least 3 statute miles, and the drone must maintain 500 feet below clouds and 2,000 feet horizontal distance from clouds. Drones cannot fly over people outside of restricted categories, and cannot drop cargo or objects except in certain controlled conditions.

KEY FACTS

- 400 feet AGL maximum altitude
- 100 mph maximum airspeed
- 3 statute mile visibility minimum
- 55 pounds maximum weight

Remote Pilot Certification

To legally operate a drone commercially under Part 107, a person must obtain a Remote Pilot Certificate from the FAA. This requires passing the Part 107 Knowledge Test (also called the FAA written exam), which consists of 60 multiple-choice questions covering regulations, airspace, weather, and operations. A passing score of 70% or higher (42 correct answers) is required. The test takes 120 minutes (2 hours) and must be taken at an FAA-approved testing center or through approved remote proctoring. The certificate is valid for 3 years. To maintain the certificate, pilots must complete FREE online recurrent training through FAA Safety Team (WINGS

or [FAASafety.gov](https://www.faa.gov/safety)) every 24 calendar months—no exam is required. Pilots must be at least 16 years old and be able to read, write, and understand English.

KEY FACTS

- 60-question knowledge test
- 70% passing score required
- Certificate valid 3 years
- Online recurrent training every 24 months (FREE, no exam)
- 16 years minimum age

Key Operational Rules

Part 107 pilots have specific responsibilities during operations. The remote pilot in command is solely responsible for the safe operation and must have a valid certificate in their possession. A visual observer (separate from the pilot) is required during operations to maintain VLOS and scan for traffic. Drones cannot be operated under the influence of alcohol (8-hour minimum between drinking and flying) or drugs that impair judgment. The aircraft must be in safe working condition, verified through preflight inspection. Operations are restricted to daylight (30 minutes after sunrise to 30 minutes before sunset) without special authorization. Preflight briefing must include weather, airspace checks (using LAANC), NOTAMs, TFRs, and emergency procedures. All crew members must understand their roles and responsibilities.

KEY FACTS

- Visual observer required, separate from pilot
- Only one remote pilot can be in command
- Daylight operations only without waiver
- Preflight inspection required
- 8-hour minimum after alcohol

Airspace and Authorization

Part 107 operations require proper airspace authorization. In controlled airspace (Class B, C, D, and some Class E areas), LAANC authorization is required. LAANC is a system that provides real-time airspace authorization for drone operations below 400 feet, allowing quick approval for operations that are compatible with manned aircraft. In uncontrolled airspace (Class G), operations may not require LAANC but must still follow all Part 107 rules. Part 107 drones cannot operate in Restricted or Prohibited Areas without waiver authorization. All pilots must check for Temporary Flight Restrictions (TFRs) before every flight, which may restrict operations during major events, emergency response, or security issues. Military Training Routes and Special Use Airspace require careful consideration.

KEY FACTS

- LAANC required in controlled airspace
- Check TFRs before every flight
- Cannot fly in Prohibited/Restricted areas
- Class G airspace may be unrestricted
- Airspace authorization takes 5-10 minutes

WEATHER

Chapter 2 **Weather Fundamentals**

Weather Elements and Flight Safety

Weather significantly affects drone operations. The primary weather elements affecting flight are wind, precipitation, visibility, clouds, temperature, and pressure. Wind affects the drone's ability to maintain position and can exceed the aircraft's capabilities in extreme conditions. Most Part 107 drones can handle winds up to 20-25 mph, though it should be verified for your specific aircraft. Precipitation (rain, snow, sleet) reduces visibility and can damage electronics if the drone is not water-resistant. Visibility (how far you can see) must be at least 3 statute miles. Clouds affect ceiling altitude available for operations; drones must stay 500 feet below clouds. Temperature affects battery performance and air density. Low temperatures reduce battery efficiency and increase power draw. Barometric pressure affects altitude calculations. Understanding how each element affects your specific aircraft is essential for safe flight planning.

KEY FACTS

- Wind affects control and position
- Rain reduces visibility and damages drones
- Visibility minimum: 3 statute miles
- Stay 500 feet below clouds
- Cold reduces battery performance

Wind and Its Effects

Wind is one of the most important weather factors for Part 107 operations. Headwind (wind from ahead) helps the drone by increasing effective airspeed and reducing ground speed. Tailwind (wind from behind) reduces effective airspeed and increases ground speed. Crosswind (wind from the side) pushes the drone laterally and must be compensated. Strong winds reduce flight endurance because the drone must work harder to maintain position. The drone's maximum wind capability depends on its design and power. Most consumer drones can handle sustained winds of 15-20 mph but not gusts beyond that. Wind direction and speed are reported from meteorological reports (METARs). Wind is reported as direction true north followed by speed. For example, "27008" means wind from 270 degrees (west) at 8 knots. Gusts are indicated by "G" notation, e.g., "27008G15" means gusting to 15 knots.

KEY FACTS

- Headwind increases control authority
- Tailwind reduces flight time
- Crosswind causes drift
- Wind from surface to 400 feet varies
- Check METAR for wind data

Visibility and Atmospheric Conditions

Visibility is the farthest distance at which objects can be clearly seen. For Part 107 operations, minimum visibility is 3 statute miles (3SM). Visibility can be reduced by several atmospheric phenomena: fog reduces visibility to less than 1/2 SM (depicted as "FG" in METARs), mist/haze reduces visibility to 1/2-6 SM ("BR" in METARs), precipitation reduces visibility (rain "RA", snow "SN"), dust or smoke ("DU", "FU") reduce visibility, and blowing snow/sand reduce visibility ("BLSN", "BLDU"). Reading METARs allows determination of current visibility. If visibility is below 3 statute miles, Part 107 operations cannot proceed. Visibility changes throughout the day - it is often poor in early morning and evening due to fog. Visibility improves during the day as sun heats the surface. Before every flight, check current visibility from METAR data. Also check TAF forecasts for any visibility changes during the planned operation time.

KEY FACTS

- Visibility minimum: 3 statute miles
- Fog reduces to <1/2 SM
- Mist/haze: 1/2-6 SM
- Check METAR for current visibility
- TAF shows forecast changes

Clouds and Ceiling Altitude

Clouds are important for two reasons: they limit the altitude available for operations and they indicate weather. Part 107 drones must stay 500 feet below clouds. Cloud height is reported in METARs using codes. "FEW" means 1-2 eighths of sky covered, "SCT" (Scattered) means 3-4 eighths, "BKN" (Broken) means 5-7 eighths, and "OVC" (Overcast) means 8-8 eighths (100% coverage). Each cloud layer is reported with an altitude. For example, "OVC025" means overcast at 2,500 feet. A ceiling is defined as the lowest layer that is broken or overcast (BKN or OVC). If the ceiling is at 2,000 feet AGL, the maximum safe operating altitude for Part 107 is 1,500 feet AGL. Since Part 107 limit is 400 feet AGL, ceiling is usually not a limiting factor for Part 107 drones. However, visibility often deteriorates near clouds, which can affect safe operations.

KEY FACTS

- Clouds reported in eighths
- FEW=1-2/8, SCT=3-4/8, BKN=5-7/8, OVC=8/8
- Stay 500 feet below
- Ceiling = lowest BKN/OVC layer
- Low clouds = poor visibility

PERFORMANCE

Chapter 3

Density Altitude & Performance**What is Density Altitude?**

Density altitude is the altitude at which an aircraft performs as if it were operating at that altitude. It combines the effects of actual altitude, temperature, and humidity into a single value. An aircraft operating on a hot, humid day at sea level performs as if it were operating at 2,000 feet or higher due to reduced air density. Density altitude is calculated using pressure altitude (actual altitude adjusted for barometric pressure) and temperature. High density altitude means thin air and reduced performance. Low density altitude means thick air and good performance. Understanding density altitude is essential for predicting aircraft performance. Drones at 400 feet AGL are typically in air with sufficient density for normal operation, but extreme conditions can still affect performance significantly.

KEY FACTS

- Combines altitude, temperature, humidity
- Hot, humid day = high DA
- Cold day = low DA
- Affects aircraft performance
- Calculate from temperature and pressure

Factors Affecting Density Altitude

Four factors affect density altitude: (1) Altitude: higher actual altitude means thinner air and higher density altitude. An airport at 5,000 feet elevation automatically has higher density altitude. (2) Temperature: higher temperatures reduce air density (warm air is less dense). A hot day significantly increases density altitude. (3) Humidity: moisture in air affects density. Surprisingly, moist air is slightly less dense than dry air at the same pressure. Very humid air slightly increases density altitude. (4) Barometric Pressure: high pressure increases density and lowers density altitude. Low pressure reduces density and raises density altitude. A combination of high altitude, high temperature, high humidity, and low pressure creates extremely high density altitude with severely reduced performance. Cold, low-altitude, dry, high-pressure days provide the best performance conditions.

KEY FACTS

- Higher altitude = higher DA
- Higher temperature = higher DA
- Higher humidity = slightly higher DA
- Higher pressure = lower DA
- Multiple factors combine

Performance Effects and Calculations

High density altitude reduces climb rate, extends takeoff distance, reduces maximum weight capacity, and shortens flight endurance. For example, if an aircraft has a climb rate of 1,000 feet per minute at sea level on a cool day, that same aircraft might have only 500 fpm climb rate on a hot day at high altitude. Density altitude is not a real altitude but an effective one based on air density. Online calculators and charts allow pilots to compute density altitude from temperature, pressure altitude, and humidity. The formula involves converting pressure altitude to density altitude using a temperature correction. A typical example: 3,000 feet pressure altitude, 30°C temperature, and reasonable humidity might yield 5,000 feet density altitude. The aircraft then performs as if flying at 5,000 feet. For Part 107 operations at 400 feet AGL, density altitude is less critical since you're operating at low altitude anyway. However, on extremely hot days, performance may still be affected.

KEY FACTS

- High DA reduces performance
- DA calculator online available
- Affects climb rate most
- Takeoff distance increases
- Hot + high alt + humid = worst

OPERATIONS

Chapter 4

Aeronautical Decision Making**What is Aeronautical Decision Making?**

Aeronautical Decision Making (ADM) is a systematic process for making decisions that affect flight safety. ADM recognizes that most accidents result from poor decisions rather than technical failures. The goal of ADM is to make the best possible decision given available information, time constraints, and resources. ADM applies to all flight planning and operational decisions. Examples include deciding to conduct or cancel a flight, choosing a flight route, deciding whether conditions are safe, managing in-flight problems, and emergency procedures. Good ADM requires recognizing the problem, analyzing options, selecting the best option, and implementing it. Poor ADM might skip analysis or implementation steps. The consequences of poor ADM can be catastrophic. ADM training is required for all pilots and emphasizes recognizing personal limitations and biases that lead to poor decisions.

KEY FACTS

- Systematic decision process
- Most accidents from poor decisions
- Applies to all flight decisions
- Requires analyzing options
- Part of pilot training

The DECIDE Model

The DECIDE model is a structured approach to ADM consisting of six steps: (D) Detect the problem - recognize that a situation requires a decision. Example: noticing deteriorating weather. (E) Estimate the severity - understand how critical the problem is. (C) Choose an action - identify options and select one. (I) Identify the best strategy - analyze the chosen action and confirm it's optimal. (D) Do it - implement the decision. (E) Evaluate the result - check if the decision achieved the goal. Each step is important. Skipping steps or rushing leads to poor outcomes. Example: detecting deteriorating visibility (D), estimating it will drop below 3SM within 30 minutes (E), choosing to land at nearest airport (C), planning the route and confirming fuel (I), executing the landing (D), evaluating success (E). Following DECIDE systematically improves decision quality and safety.

KEY FACTS

- Six-step structured process
- Detect problem first
- Estimate severity
- Choose action deliberately
- Evaluate result after

Risk Assessment and Mitigation

Risk assessment is fundamental to ADM. Every flight has risks: weather, mechanical, human, airspace, environmental. Good ADM quantifies risks and plans mitigation. For example, flying on a marginal weather day carries risk of weather deterioration. Mitigation might include staying near airport, planning alternate location, or canceling. Each risk should have a mitigation strategy. Some risks are mitigated through prevention (maintenance reduces mechanical risk), some through contingency planning (weather deterioration mitigation), some through training (pilot technique reduces handling risk). Risk can be reduced but not eliminated. When residual risk (after mitigation) is acceptable, proceed; when unacceptable, cancel or modify. Risk acceptance is a personal decision but must consider only acceptable risks to self and others. No flight is worth injury or death. Conservative risk assessment is a sign of good judgment.

KEY FACTS

- Identify all risks
- Develop mitigation for each
- Residual risk must be acceptable
- When unacceptable, cancel
- Conservative approach best

REGULATIONS

Drone Registration Requirements

All drones used for Part 107 operations must be registered with the FAA. Registration applies to drones weighing 0.55 pounds or more up to 55 pounds. The registration fee is \$5 and the registration is valid for 3 years. To register, the aircraft owner must be at least 13 years old (or have parent/guardian provide information) and have an FAA account (username). The registration number must be marked on the outside of the aircraft in a place that can be easily read. Drones under 0.55 pounds do not require registration but must still follow all other Part 107 rules if used commercially. The registration number serves as the aircraft identification and must be displayed clearly. Commercial operators should maintain registration documentation at the flight site.

KEY FACTS

- Registration fee: \$5
- Valid 3 years
- Applies to drones 0.55+ pounds
- Must mark registration number on aircraft
- Owner must be 13+ years old

Remote ID Requirements

Remote ID is a technology that broadcasts the location and identification of a drone. All drones must have Remote ID capability to broadcast their location, altitude, and operator identification. This allows authorities to identify the drone and operator in case of incidents. Remote ID can be broadcast via network (cellular) or broadcast (direct radio). Network-based Remote ID requires internet connectivity and broadcasts via cellular networks. Broadcast-based Remote ID transmits directly without internet. Most consumer drones now include Remote ID capability. Some limited exemptions exist for drones under 0.55 pounds, certain recreational flights, and operations in controlled airspace with specific authorization. Remote ID information is displayed publicly and can be accessed by authorized users. Part 107 operators must ensure their drones are Remote ID compliant.

KEY FACTS

- All drones must broadcast Remote ID
- Shows location and operator ID
- Network or broadcast capability
- Exemptions for aircraft under 0.55 lbs
- Enables FAA compliance tracking

Operator Registration and Tracking

Beyond aircraft registration, Part 107 operators should maintain records of their operations for regulatory compliance and safety. The remote pilot certificate itself serves as the operator authorization, but operators should keep documentation of flights including date, time, location, mission type, duration, and crew members. This documentation can be useful for insurance purposes and demonstrating compliance with regulations. Some operations may require special documentation including waivers, TFR approvals, or LAANC records. Maintaining detailed flight logs helps establish operational history and professionalism, especially important for commercial operators. Records should be kept for at least 24 months and should include any incidents or unusual occurrences during flights.

KEY FACTS

- Maintain flight logs with date, time, location
- Document all incidents
- Keep 24-month record retention
- Certificate is primary operator authorization
- Documentation aids insurance and compliance

Liability and Insurance Considerations

While Part 107 insurance is not mandated by the FAA, it is highly recommended for commercial operators. Drone liability insurance covers property damage and bodily injury caused by the aircraft. Standard policies provide \$1-2 million in coverage, which is appropriate for most commercial operations. Insurance protects against claims from property damage below (e.g., damaged roofs or vehicles), bodily injury, and third-party liability. Some operations or clients may require proof of insurance before allowing operations. The cost of insurance varies based on coverage limits, aircraft type, operational area, and claims history. Many operators find insurance essential for professionalism and risk management. Operators should understand policy exclusions and ensure coverage matches their operational needs.

KEY FACTS

- Insurance not required but highly recommended
- Standard coverage: \$1-2 million
- Covers property damage and bodily injury
- Costs vary by coverage and risk
- Professional operators carry insurance

WEATHER

METAR Format and Elements

A METAR (Meteorological Aerodrome Report) is a standardized weather report issued by airports every hour (and sometimes more frequently). METARs contain coded information about wind, visibility, precipitation, clouds, temperature, dew point, and pressure. The format is standardized internationally. A typical METAR begins with the airport identifier (e.g., "KJFK"), followed by the date and time in UTC (Z notation). The first complete element is wind: direction and speed. Following wind is visibility, usually in statute miles (SM). Then comes weather phenomena (precipitation, obscuration). Clouds are reported next, usually with multiple layers. Temperature and dew point are reported in Celsius. Finally, altimeter setting is reported (barometric pressure). Each element has specific abbreviations that must be memorized. Learning to read METARs is essential for flight planning and weather assessment.

KEY FACTS

- Hourly reports from airports
- Format: KJFK 121251Z...
- Wind first (direction/speed)
- Visibility in SM
- Multiple layers of clouds

Wind and Visibility Codes

Wind is reported as four digits for direction (010-360 degrees true north) plus two or three digits for speed. "27008KT" means wind from 270 degrees at 8 knots. Gusts are shown as "G15" appended to wind speed, meaning gusting to 15. "VRB" means variable wind (less than 3 knots). Visibility is reported in statute miles, often as a whole number or fraction. "10SM" means 10 statute miles. "3SM" means 3 statute miles minimum visibility (Part 107 requirement). Visibility can be fractional: "1/2SM" (half mile), "1/4SM" (quarter mile). When visibility is very low (less than 1/4SM), it's often reported as "RMK A02" or similar comments. Visibility reported with direction like "1/2SM NW" indicates reduced visibility in one direction. Always note exact visibility from the METAR as this determines if operations can proceed (minimum 3SM for Part 107).

KEY FACTS

- Wind: direction + speed (knots)
- Gusts appended with G notation
- Variable wind: VRB
- Visibility in statute miles
- Part 107 needs 3SM minimum

Weather Phenomena Codes

Weather phenomena are reported using two or three character codes. Intensity is indicated by prefix: "-" (light), no prefix (moderate), "+" (heavy). Precipitation: "RA" (rain), "SN" (snow), "DZ" (drizzle), "SH" (showers), "FZ" (freezing). Example: "-RA" is light rain, "+SN" is heavy snow. Obstructions to vision: "FG" (fog), "BR" (mist), "HZ" (haze), "DU" (dust), "FU" (smoke). Storm phenomena: "TS" (thunderstorm), "SQ" (squall). Water phenomena: "FZ" prefix indicates freezing conditions. Example: "FZRA" is freezing rain. Multiple phenomena can be reported. For Part 107, any weather reducing visibility below 3SM or producing precipitation should halt operations. Heavy precipitation is especially hazardous due to visibility reduction and potential aircraft damage.

KEY FACTS

- Intensity: - (light), none (moderate), + (heavy)
- RA=rain, SN=snow, DZ=drizzle
- FG=fog, BR=mist, HZ=haze
- TS=thunderstorm
- FZ prefix=freezing conditions

Clouds, Temperature, and Altimeter

Clouds are reported with coverage and altitude. "FEW030" = few at 3,000 feet, "OVC050" = overcast at 5,000 feet. Multiple cloud layers can be reported, each with altitude. Temperature and dew point are reported as "T25/15" meaning 25°C temperature and 15°C dew point. When temperature and dew point are close (small difference), moisture is high and fog/clouds likely. Altimeter setting is reported as "A2990" meaning 29.90 inches of mercury. This is the barometric pressure. Standard altimeter setting is 29.92. Higher values (A2995) indicate high pressure and good weather. Lower values (A2985) indicate low pressure and poor weather. Dew point depression (temperature minus dew point) indicates stability: small difference means moisture is high, large difference means dry air. These elements together give a complete picture of current conditions for flight planning.

KEY FACTS

- Cloud layers reported separately
- T25/15 = 25°C temp, 15°C dew point
- Small dew point spread = high moisture
- Altimeter: A2990
- A>2992 = high pressure
- A<2992 = low pressure

PERFORMANCE

Weight and Balance Principles

Weight and balance is critical for safe aircraft operation. The aircraft's center of gravity (CG) must be within the allowable range for safe flight. If CG is too far forward, the aircraft becomes nose-heavy and requires excessive down elevator to maintain level flight. If CG is too far aft (rear), the aircraft becomes tail-heavy and is unstable or uncontrollable. Every aircraft has a CG envelope (range of allowable CG positions). The CG envelope is defined by forward and aft limits. Aircraft weight must also be within limits. Maximum takeoff weight (MTOW) for Part 107 drones is 55 pounds. Exceeding MTOW is a Part 107 violation. The aircraft's empty weight plus all equipment, batteries, and payload must not exceed 55 pounds. Weight distribution affects the CG position. Moving weight forward (toward the nose) moves CG forward. Moving weight aft (toward the tail) moves CG aft.

KEY FACTS

- CG must be in envelope
- CG too far forward = nose-heavy
- CG too far aft = unstable
- MTOW = 55 pounds
- Part 107 weight limit cannot be waived

Calculating Center of Gravity

CG is calculated using the moment principle. Moment equals weight times distance from a reference point. For example, a 10-pound package 24 inches from the reference point creates a moment of 240 pound-inches. The aircraft's total moment is the sum of all component moments. CG location equals total moment divided by total weight. For example, total moment of 3,000 pound-inches and total weight of 50 pounds yields CG at 60 inches from reference. Each aircraft has an empty weight and empty CG printed in aircraft documentation. Adding payload weight and calculating the new moment gives the loaded CG. As long as the loaded CG is between the forward and aft limits, the aircraft is safe to fly. Example: empty weight 45 lbs at 62 inches CG (moment = 2,790). Adding 5-lb camera at 70 inches from reference (moment = 350). Total weight = 50 lbs, total moment = 3,140, new CG = 62.8 inches.

KEY FACTS

- Moment = weight × distance
- CG = total moment ÷ total weight
- Aircraft CG envelope given
- Calculate loaded CG before flight
- Must be in allowable range

Loading and Payload Considerations

Payload (equipment beyond basic aircraft) affects both weight and CG. A camera mounted on the front of the aircraft adds weight forward, moving CG forward. A battery in the rear adds weight aft, moving CG rearward. The location of payload is as important as its weight. A light camera on the front can move CG further than a heavy battery in the rear if positioned appropriately. Unbalanced loading creates CG outside the envelope. For example, if you load a camera but no rear weight, CG might move too far forward and become nose-heavy. The solution is to load batteries rearward or position the camera differently. Before adding any payload, verify that the aircraft will remain in CG envelope. Software tools can help calculate CG with various loading configurations. Testing is important: with unfamiliar loading, perform test flights at altitude with margin for error before committing to operations. Never intentionally load the aircraft unbalanced assuming you can compensate with control inputs - this reduces stability and performance.

KEY FACTS

- Payload location affects CG
- Front load = CG forward
- Rear load = CG aft
- Calculate CG with all payload
- Test unfamiliar configurations

OPERATIONS

Chapter 8

Crew Resource Management**What is Crew Resource Management?**

Crew Resource Management (CRM) is the effective use of all available resources (people, equipment, information) to conduct safe operations. CRM recognizes that accidents often result from poor coordination, communication, or decision-making among crew members rather than technical failures. Even with only a remote pilot and visual observer, CRM principles apply. The remote pilot is the pilot in command with ultimate responsibility. The visual observer is an essential crew member with specific responsibilities. Both must communicate effectively, coordinate actions, and support each other. CRM emphasizes that the pilot must welcome input from crew members and listen to concerns. A visual observer who sees a safety problem must speak up, and the pilot must listen and act. CRM also recognizes human limitations like fatigue, stress, and distraction, and develops procedures to manage these factors.

KEY FACTS

- Uses all available resources
- Prevents miscommunication
- Pilot welcomes crew input
- Crew speaks up about concerns
- Manages human limitations

Communication and Coordination

Effective communication is the cornerstone of CRM. Clear, concise, two-way communication prevents misunderstandings. The remote pilot should clearly state the operation plan, assign crew roles, establish communication procedures. The visual observer should confirm understanding and report observations regularly. During flight, the pilot should provide continuous information about aircraft status, altitude, battery, time remaining. The observer should report traffic, hazards, wind conditions, and any concerns. Information flow must be bidirectional - not just pilot to observer but observer to pilot as well. Standardized terminology reduces confusion. For example, specific call-outs for altitude changes, emergency procedures, or hazard reports. Closed-loop communication ensures information is received: pilot says "landing in 2 minutes," observer acknowledges "landing in 2 minutes, understood." Common mistakes include inadequate briefing, poor radio discipline, failure to acknowledge information, and crew members not speaking up about concerns.

KEY FACTS

- Two-way communication essential
- Briefing before flight
- Continuous information exchange
- Standardized terminology

- Acknowledge information received

Error Management and Challenge

Errors happen - pilots make mistakes. CRM goal is catching errors before they cause accidents. Good CRM includes cross-checking: the observer verifies what the pilot reports. If the observer sees something different, they speak up. Example: pilot says "altitude 350 feet" but observer sees 425 feet - observer must announce the discrepancy. The pilot then checks instruments and corrects if needed. Challenge and response procedures formalize this: if something seems wrong, anyone can challenge. The person being challenged must explain or correct. This removes hierarchy from error management - even the visual observer can challenge the pilot if something seems unsafe. Everyone is responsible for safety. Error management also includes forgiveness: if the observer catches an error, the goal is correction, not blame. A culture of no-blame encourages crew members to speak up without fear of punishment.

KEY FACTS

- Errors will happen
- Catch before accidents occur
- Cross-check information
- Challenge/response procedures
- No-blame culture

REGULATIONS

What is a Part 107 Waiver?

A waiver is an FAA authorization that allows operations to deviate from certain Part 107 rules when the FAA determines the operation can be conducted safely. Waivers are not automatically granted and must be specifically requested through the FAA. The FAA uses a risk-based analysis to determine whether to approve a waiver. Waivers are typically issued for specific operations and have a defined validity period, usually 2 years. Some Part 107 rules cannot be waived under any circumstances: the 55-pound weight limit, the 400-foot altitude limit, the 100 mph airspeed limit, and the requirement for remote pilot certification. Waivers might be requested for operations like flying beyond visual line of sight (BVLOS), night operations, flying over people, operations in special use airspace, or other activities that would normally violate Part 107.

KEY FACTS
• Authorization to deviate from Part 107 rules • FAA uses risk-based analysis • Issued for specific operations • 55 lbs, 400 ft, 100 mph cannot be waived • Typically valid 2 years

Requesting a Waiver

To request a Part 107 waiver, operators must submit an application through the FAA's online system (DroneZone or similar platform). The application requires detailed information about the proposed operation including the specific rules to be waived, the location, duration, aircraft specifications, safety mitigation measures, and operational procedures. The FAA recommends submitting waiver requests at least 90 days before the intended operation, though emergency waivers may be available on shorter notice. The application should demonstrate that the operation can be conducted safely despite deviating from standard Part 107 limits. This might include additional safety measures like extra spotters, insurance requirements, weather minimums, or operational restrictions. The FAA may approve, deny, or approve with conditions. Operators should be prepared to justify every aspect of their operation.

KEY FACTS
• Submit through FAA online system • 90-day advance notice recommended • Must justify safety measures • FAA may add conditions to approval • Approval is not guaranteed

Common Waiver Categories

Note: Night operations NO LONGER require a waiver since the April 2021 rule change — they just need anti-collision lighting. The most commonly requested waivers are now: BVLOS (Beyond Visual Line of Sight) operations for long-range surveying and mapping, requiring additional spotters, communication systems, and tracking technology. Operations Over People waivers allow flights over crowds or populated areas with specific aircraft design requirements and operational controls. Operations from moving vehicles. Flights above 400 feet AGL. Multiple aircraft operations by a single pilot. Each waiver type has specific safety requirements and mitigation measures that must be demonstrated to the FAA.

KEY FACTS

- Night ops NO LONGER need a waiver (since 2021)
- BVLOS is most common waiver now
- Over-people has 4 categories (some no waiver needed)
- Cannot waive the 55-pound weight limit
- Must demonstrate equivalent safety level

Waiver Conditions and Compliance

When a waiver is approved, the FAA includes specific conditions that must be followed during operation. Violating waiver conditions is a serious violation and can result in certificate revocation and significant penalties. Operators must document compliance with waiver conditions, including briefing all crew members on the conditions. Conditions might include altitude restrictions, weather minimums, notification requirements, escort aircraft presence, or specific operational procedures. Some waivers require notice to ATC before operations. Operators should thoroughly understand all waiver conditions and communicate them to the entire crew. If conditions cannot be met at the time of operation, the flight should not proceed even if the waiver is valid. Waivers should be printed and available during the operation. Maintaining records of waiver compliance demonstrates professionalism and aids in future waiver requests.

KEY FACTS

- Waiver conditions are mandatory
- Brief all crew on conditions
- Violating conditions = serious violation
- Document compliance
- Carry printed waiver at flight site

WEATHER

TAF Format and Purpose

A TAF (Terminal Aerodrome Forecast) is an airport weather forecast issued 4-5 times daily. TAFs are valid for 24 hours for regular airports and 30 hours for selected airports. The TAF provides a detailed forecast of expected weather changes during the forecast period. The format is similar to METAR but with timing information. A TAF begins with "TAF" or "TAFL" (light version), then airport ID, issuance date/time, and valid period. The initial group covers the first period (usually 6-9 hours) without changes predicted. Then change groups follow, indicated by "FM" (from), "BECMG" (becoming), or "TEMPO" (temporary). "FM" indicates a new forecast period starting at a specific time. "BECMG" indicates gradual changes within 1-2 hours. "TEMPO" indicates temporary conditions lasting less than 1 hour. Understanding TAF timing is essential for predicting weather during flight operations.

KEY FACTS

- Issued 4-5 times daily
- Valid 24-30 hours
- Format similar to METAR
- FM = new period
- BECMG = gradual change
- TEMPO = temporary

TAF Interpretation and Change Groups

The TAF base forecast describes expected weather from issuance through a specified time (usually 6-9 hours). This includes wind, visibility, weather, and clouds. Change groups follow with specific timing. "FM" indicates "from" a specific time and shows the new forecast. For example, "FM141800" means "from the 14th at 1800Z" with the following conditions. Multiple FM groups may be present as weather evolves. "BECMG" indicates conditions are becoming (changing) within a 1-2 hour window. Written as "BECMG 1418" means "becoming between 1400Z and 1800Z." The forecasted condition should be achieved by the end of the time window. "TEMPO" indicates temporary conditions, usually weather. Written as "TEMPO 1920" means temporary conditions between 1900Z and 2000Z. These temporary conditions may repeat but will not be continuous. TEMPO conditions typically last less than 1 hour when they occur.

KEY FACTS

- Base forecast initial conditions
- FM = new forecast period
- BECMG = gradual change 1-2 hrs
- TEMPO = temporary conditions

- Time format: DDHR (day/time UTC)

Using TAFs for Flight Planning

TAFs help predict weather conditions during planned operations. Before requesting a flight, check the TAF for your airport. Note any FM (from) times when conditions change. If a significant change is forecast to occur during your planned operation time, consider adjusting your time or operation. Check BECMG and TEMPO groups carefully: if your operation conflicts with periods of poor weather (low visibility, storms, strong winds), reschedule if possible. TAFs are forecasts and not guaranteed - actual conditions may differ. Even if TAF shows good conditions, always check current METARs before launching. Comparing the most recent METAR with the TAF helps verify forecast accuracy. If conditions are developing faster than forecast, cancel the operation. TAF skill develops with practice and comparison to actual outcomes. Experienced pilots learn to recognize forecast accuracy and unreliability.

KEY FACTS

- Check for FM changes during operation time
- Note BECMG time windows
- TEMPO may cause operation delays
- TAFs are forecasts only
- Verify with current METAR before launch

TAF Code Examples

A complete TAF example: "KJFK 121251Z 1212/1318 27008KT 10SM FEW250 FM131500 31010KT 10SM OVC200". This means: KJFK airport, issued 12th at 1251Z, valid 12th 1200Z through 13th 1800Z. Initial forecast: wind 270 at 8 knots, 10SM visibility, few clouds at 25,000 feet. FM131500: From 13th at 1500Z, wind 310 at 10 knots, 10SM visibility, overcast at 20,000 feet. Another example: "KORD 151120Z 1512/1618 32012G20KT 8SM -RA BKN040 TEMPO 1512/1614 3SM +RA". KORD airport, valid 15th 1200Z through 16th 1800Z. Wind 320 at 12 knots gusting to 20, 8SM visibility, light rain, broken clouds 4,000 feet. TEMPO: temporarily 3SM visibility with heavy rain between 1200Z and 1400Z. Decoding requires careful reading of abbreviations and time notations.

KEY FACTS

- Wind: direction and speed
- Visibility always included
- Cloud layer with altitude
- FM indicates time change
- TEMPO indicates brief poor conditions

OPERATIONS

Chapter 11

Physiology: Alcohol, Drugs & Fatigue**Alcohol and Flight Safety**

Alcohol impairs judgment, coordination, vision, and reaction time even in small amounts. A single drink can measurably reduce pilot performance. Alcohol affects the brain's ability to process information and make decisions. It reduces situational awareness and increases risk-taking. The FAA prohibits flying with a blood alcohol content (BAC) of 0.04% or higher. For context, a 170-pound person reaches 0.04% BAC after about 2-3 drinks. Alcohol is absorbed slowly (about one drink per hour is metabolized), so a pilot who drank the night before might still have residual alcohol. The 8-hour "bottle to throttle" rule provides margin: don't fly within 8 hours of consuming alcohol. This allows most alcohol to be metabolized while providing safety margin. Even with no detectable BAC, impaired judgment from recent drinking is a concern. Pilots must never fly after drinking or if intoxicated. Drinking during operations is strictly prohibited.

KEY FACTS

- Impairs judgment and coordination
- BAC limit: 0.04%
- 8-hour rule from drinking
- One drink per hour metabolizes
- Never fly while intoxicated

Drugs and Medications

Any drug affecting judgment, coordination, or consciousness is prohibited before flying. This includes marijuana (even in states where legal), cocaine, amphetamines, depressants, and hallucinogens. Even "recreational" use disqualifies flight. Prescription medications require caution. Some medications are safe to fly with (like blood pressure medication), while others are not (sedatives, painkillers with narcotics). Pilots taking any medication should consult the FAA's Medications Database or their Aviation Medical Examiner before flying. Over-the-counter medications can also cause problems: cold medicines with decongestants, antihistamines, and cough medicines can cause drowsiness or impair judgment. Combination medications (cold + pain relief) might contain problematic ingredients. If any medication could affect performance, don't fly. The general rule: anything that impairs judgment or physical capability is prohibited. When in doubt, ask the FAA or skip the flight.

KEY FACTS

- Illegal drugs prohibited
- Marijuana prohibited
- Prescription meds require check
- OTC meds can be problematic
- Impairment = not flying

Fatigue and Alertness

Fatigue impairs decision-making and reaction time similar to alcohol. A tired pilot makes poor decisions and reacts slowly to problems. Fatigue can be acute (from insufficient sleep) or chronic (from sustained operations). Acute fatigue from one poor night's sleep significantly affects performance. Chronic fatigue from multiple days of inadequate sleep causes cumulative impairment. Circadian rhythm (body's internal clock) affects fatigue: flying during hours when you normally sleep causes fatigue even if you slept 8 hours the day before. Early morning flights require very good sleep the previous night. Flying in afternoon after poor sleep is dangerous. The minimum sleep recommendation is 8 hours the night before flying. For early flights, sleep should be even longer. A tired pilot should not fly. Fatigue is as disabling as alcohol or drugs but is often self-imposed. Taking breaks during long operations helps manage fatigue. Multi-person crews (with visual observer taking breaks) helps manage fatigue better than single-pilot operations.

KEY FACTS

- Fatigue = impaired judgment
- One bad night affects performance
- Circadian rhythm important
- 8 hours sleep minimum
- Tired pilot = don't fly

REGULATIONS

Night Operations Rules

Since the April 2021 rule update, Part 107 operations at night are permitted **WITHOUT** a waiver under two conditions: (1) the aircraft must have anti-collision lighting visible for at least 3 statute miles with a flash rate sufficient to avoid collision, and (2) the remote pilot must have completed initial or recurrent knowledge testing after April 6, 2021. Daylight operations are defined as 30 minutes before official sunrise to 30 minutes after official sunset. Civil twilight (the 30-minute windows around sunrise/sunset) is considered part of the allowed daylight period. Night operations still present unique challenges including difficulty seeing the aircraft, reduced visual observer effectiveness, and obstacle avoidance challenges.

KEY FACTS

- Night ops allowed **WITHOUT** waiver since April 2021
- Anti-collision lighting must be visible for 3+ statute miles
- Pilot must have completed training after April 6, 2021
- Daylight = 30 min before sunrise to 30 min after sunset
- Civil twilight is included in daylight hours

Night Flight Considerations

Night flying introduces special hazards and requires additional safety measures. Without visual reference to the ground, spatial disorientation is more likely. The visual observer's effectiveness is greatly reduced at night. Lighting from ground sources (streetlights, buildings) can create disorientation and make it difficult to judge the aircraft's altitude and position. Collision hazards increase as other aircraft may be harder to see. Obstacles that are easily visible during day (trees, power lines, towers) are invisible at night. Night operations typically require additional spotters positioned to watch specific areas or warn of hazards. Communication systems may need to be enhanced to coordinate between multiple spotters. Weather deterioration can occur rapidly at night and is harder to detect. All crew members should understand night-specific hazards and have practiced night operations in training.

KEY FACTS

- Spatial disorientation risk increases
- Visual observer less effective
- Ground lighting can cause disorientation
- Obstacles invisible at night
- Additional spotters recommended

Operations Over People - Categories and Requirements

Part 107 allows operations over people under four specific categories based on aircraft characteristics. Category 1 (minimal hazard): Aircraft weighing under 0.55 pounds including all attachments. No special authorization needed. Category 2 (low risk): Aircraft with no exposed rotating parts that could lacerate skin, tested per specific design standards. Category 3 (outdoor assemblies): Similar to Category 2 but designed for operations over open-air assemblies of people. Requires Remote ID broadcast capability. Category 4 (high-risk operations): FAA airworthiness-certified aircraft. Requires Remote ID and speed limited to 50 mph. Each category has specific aircraft design requirements, Remote ID requirements in some cases, and operational procedures. Aircraft must be tested and certified to the appropriate category. Operations must follow all applicable restrictions and mitigation measures.

KEY FACTS

- Category 1: under 0.55 lbs, no auth needed
- Category 2: no sharp/rotating parts, design tested
- Category 3: Cat 2 for assemblies, needs Remote ID
- Category 4: airworthiness certified, Remote ID + 50mph limit
- Aircraft characteristics determine category

Over-People Operations Safety and Mitigation

Operations over people require extensive safety planning and mitigation measures. A detailed risk assessment must identify all hazards and how they will be mitigated. Insurance is essential and typically required as a condition of approval. Multiple spotters should be positioned to monitor the operation and warn of hazards. Communication protocols must be established for quick problem reporting. Contingency procedures must be planned for aircraft malfunction, weather changes, or other emergencies. A safety perimeter should be established to keep unauthorized people away from the operation area. The operation should be briefed thoroughly to all crew members, including spotters and any site coordinators. Aircraft should be tested and confirmed airworthy before operations. Wind and weather conditions should be continuously monitored. If conditions change or problems develop, the operation should be suspended immediately. Documentation of the operation, including weather, crew briefing, and any incidents, should be maintained.

KEY FACTS

- Detailed risk assessment required
- Insurance essential
- Multiple spotters needed
- Communication protocols established
- Contingency procedures for emergencies

WEATHER

Thunderstorm Formation and Stages

Thunderstorms are convective storms that form in unstable air with moisture and lifting. Three stages define a thunderstorm: Developing stage (cumulus): updrafts form as warm air rises, creating the cloud structure. Light precipitation may begin. Mature stage: both updrafts and downdrafts present, heavy precipitation at peak intensity, strong winds, possible hail. This is the most dangerous stage. Dissipating stage: downdrafts dominate, updrafts weaken, precipitation decreases, but wind shear remains hazardous. Strong thunderstorms develop when the atmosphere is unstable (temperature decreases with altitude), moisture is present, and lifting mechanism exists (fronts, mountains, solar heating). Supercell thunderstorms are particularly dangerous, rotating storms that can last hours and produce tornadoes. Part 107 drones should stay at least 10 nautical miles away from thunderstorms. Thunderstorms are hazardous due to severe turbulence, wind shear, hail, lightning, and other severe weather.

KEY FACTS

- Developing, Mature, Dissipating stages
- Mature stage most dangerous
- Requires instability, moisture, lifting
- Stay 10nm from thunderstorms
- Supercells produce tornadoes

Hazards Within Thunderstorms

Severe turbulence in thunderstorms can cause structural damage to aircraft. Large hail can damage or destroy aircraft. Lightning can damage electrical systems and is lethal. Wind shear (rapid wind direction/speed changes) can cause loss of control. Downbursts and microbursts (strong downward wind currents) create low-level wind shear. These phenomena make thunderstorms the most dangerous weather phenomenon for aviation. A downdraft can have wind speeds exceeding 100 mph downward, creating extreme hazard. Gust fronts (boundaries where downdraft cool air meets warmer air) create sudden wind changes. The greatest danger from thunderstorms is microbursts, which can reduce lift suddenly and cause loss of control. Tornadoes can form from supercell thunderstorms and create catastrophic wind damage. Most weather-related accidents involve thunderstorms or conditions associated with them.

KEY FACTS

- Severe turbulence causes damage
- Hail damages aircraft
- Lightning is lethal
- Wind shear causes loss of control

- Microbursts = extreme downwind

Identification and Avoidance

Thunderstorms are identified by radar (visible on weather graphics), METAR codes (TS for thunderstorm), TAF forecasts (showing storm probability), and visual observation. Dark, towering clouds with a dark base indicate storms. Lightning is a positive identification. METARs showing "TS" or "+RA" (heavy rain) indicate storm activity. TAFs forecast storm probability and timing. Weather apps and websites show storm locations and movement. Radar is available through aviation weather services. The key to thunderstorm safety is avoidance. Never intentionally fly through or near thunderstorms. Do not rely on small gaps between storms as routes - storms can intensify and expand. If caught unexpectedly near a storm, move away immediately. The rule of thumb: stay 10 nautical miles away horizontally from thunderstorms and above 20,000 feet (not applicable to Part 107 at 400 feet). For Part 107, the best strategy is checking forecasts and avoiding areas where thunderstorms are predicted.

KEY FACTS

- TS in METAR indicates activity
- Dark towering clouds indicate storms
- Lightning = positive identification
- 10nm horizontal separation
- Avoid storm areas entirely

Other Hazardous Weather Phenomena

Wind shear can occur without thunderstorms, particularly on approaches to airports during temperature inversions. Wind shear causes rapid changes in airspeed and can cause loss of control, especially at low altitude. Icing occurs when flying through clouds with supercooled water droplets. At 400 feet AGL, icing is rare but possible in winter. Avoid flying in visible moisture (clouds) in cold temperatures. Low-level wind shear can occur in areas with complex terrain (mountains, valleys) or near the surface during frontal passages. Mountain wave turbulence occurs on the lee side of mountains as air flows over them. This turbulence can extend from the surface to high altitudes. Fog reduces visibility below safe minimums and cannot be avoided by altitude change (at 400 feet). Dust storms ("haboobs") can reduce visibility to zero within seconds. Sand and dust storms are extremely dangerous and should be avoided entirely. The key to avoiding hazardous weather is checking forecasts and having contingency plans.

KEY FACTS

- Wind shear on approaches
- Icing possible in cold clouds
- Mountain wave on lee slopes
- Fog reduces visibility to zero

- Dust storms = zero visibility

OPERATIONS

Chapter 14 **Emergency Procedures**

What Constitutes an Emergency?

An emergency is any situation requiring immediate action to ensure safety. Examples include: loss of control due to wind or malfunction, loss of communication with the aircraft, loss of battery/power leading to inability to control descent, aircraft malfunction affecting flight characteristics, encounter with manned aircraft requiring immediate evasion, weather deterioration requiring immediate descent or landing, loss of visual line of sight causing inability to control the aircraft. Emergencies require immediate response with priority shifted from completing the mission to maintaining safety. In many cases, landing immediately (even in non-ideal location) is the correct response. Emergency authority allows deviation from Part 107 rules if necessary to ensure safety. However, emergency authority is not permission to violate rules without justification - it's a legal protection if safety truly requires rule deviation.

KEY FACTS

- Loss of control = emergency
- Comm loss = emergency
- Battery critical = emergency
- Weather deterioration = emergency
- Immediate action required

Emergency Procedures and Recovery

Emergency procedures vary by aircraft and failure type. General principles: (1) stabilize the aircraft - reduce inputs, allow system stabilization; (2) identify the problem; (3) apply known recovery procedure; (4) if unknown, try basic controls (throttle, trim, control inputs); (5) land safely if unable to recover. For loss of control: reduce control inputs, center all controls, allow aircraft to stabilize, then small inputs to correct. For battery failure: land immediately at safe location. For wind gust loss of control: lower altitude if safe to do so (stronger winds at height), find sheltered area, land. For communication loss: return to starting position or land immediately. Contingency planning before flight helps - know where safe landing areas are, have return-to-home function programmed, know procedures for your specific aircraft. Never continue attempting to fly with a serious problem. Land immediately and assess.

KEY FACTS

- Stabilize aircraft first
- Identify problem second
- Apply recovery procedure
- Land if unable to recover
- Contingency plan before flight

Emergency Equipment and Resources

Emergency equipment should be available at the flight site: first aid kit for injuries, fire extinguisher for battery fires, spare batteries if mission-critical, landing gear or parachute if aircraft equipped, communication device (phone) to call emergency services. For battery emergencies, always have a safe storage location away from people and fuel. Lithium batteries can catch fire if damaged. A metal container filled with sand is traditional but modern practices include specialized battery storage bags. Emergency contact information should be available: phone numbers for local ATC, Flight Service, emergency responders. Some operations require emergency response plans and briefing. For operations over people, emergency procedures must be extensively planned. Know the location of nearest hospitals for serious injuries. Emergency procedures should be briefed to all crew before every flight. Everyone must know their role in an emergency. The visual observer should know to immediately alert the pilot of any problems.

KEY FACTS

- First aid kit available
- Fire extinguisher for batteries
- Spare batteries for critical ops
- Emergency contacts listed
- Crew briefing required

AIRSPACE

Seven Classes of Airspace

The National Airspace System is divided into seven classes: A, B, C, D, E, F (not commonly used in US), and G. Classes A through E are controlled airspace where air traffic control (ATC) services are provided. Class G is uncontrolled airspace where no ATC service is available. Class A airspace (18,000 feet MSL and above) is used for high-altitude instrument flight and is not relevant to Part 107 operations. Part 107 drones operate at 400 feet AGL, which is in the lower portions of the airspace system. The classification of airspace at a specific location depends on proximity to airports, navigation aids, and population density. Understanding the airspace class at your intended operation location is critical for determining what authorizations are needed. Class B, C, D are defined by airports; Class E can vary; Class G is the default when no controlled airspace exists.

KEY FACTS

- Classes A-E are controlled airspace
- Class G is uncontrolled airspace
- Part 107 limited to 400 feet AGL
- Classification varies by location
- Different requirements per class

Controlled vs. Uncontrolled Airspace

Controlled airspace is where ATC services are provided and pilots must follow specific rules. In controlled airspace, ATC monitors all aircraft, provides separation services, and issues clearances for operations. Controlled airspace includes Class A (high altitude), B (busy airports), C (medium airports), D (smaller airports with towers), and E (general instrument approach areas). Uncontrolled airspace is Class G where no ATC services are provided. Pilots have more freedom in Class G but must follow rules of right-of-way and see-and-avoid. For Part 107 operations, controlled airspace requires LAANC authorization before flight. Uncontrolled airspace may not require LAANC in some areas, but Part 107 rules still apply. The distinction between controlled and uncontrolled affects what authorizations are needed and what procedures must be followed.

KEY FACTS

- Controlled airspace: A, B, C, D, E
- Uncontrolled: Class G
- ATC services in controlled airspace
- LAANC required in controlled airspace
- Part 107 rules apply everywhere

Airspace Dimensions and Altitudes

Airspace is defined by horizontal and vertical boundaries. Horizontal boundaries define the lateral extent of an airspace area, often described in nautical miles from an airport or navigation aid. Vertical boundaries define the altitude limits. Altitudes are given in either Mean Sea Level (MSL) or Above Ground Level (AGL). MSL is measured from sea level, while AGL is measured from the ground elevation at that location. Part 107 altitude limits are specified in AGL. For example, Class B airspace surrounding major airports might have a 5nm radius at surface level, 10nm to 3000 feet MSL, and 20nm above 3000 feet. Class D airspace around smaller airports typically extends to 2,500 feet AGL within a 5nm radius. Understanding these dimensions helps determine when you enter or exit an airspace area. Sectional charts show airspace boundaries and altitude restrictions.

KEY FACTS

- Horizontal and vertical boundaries
- MSL vs. AGL altitudes
- Part 107 limits in AGL
- Class B extends 20-30 nm
- Sectional charts show boundaries

Authorization and Communication Requirements

Different airspace classes have different authorization and communication requirements. Class B requires ATC clearance before entry. Class C requires two-way radio communication and ATC authorization. Class D requires two-way radio communication with the tower. Class E requires ATC service but not necessarily clearance. Class G has no requirements. For Part 107 drones, which lack radios, LAANC authorization functions as the ATC clearance in controlled airspace. LAANC checks the operation against known manned aircraft activities and either authorizes or denies the request. Approval is typically quick (minutes), but authorization is never guaranteed. If denied, the operator must request alternative approaches or times. Some areas have LAANC available 24/7 while others are limited. Understanding your local airspace and available authorization methods is essential for operational planning.

KEY FACTS

- Class B requires clearance
- Class C requires communication
- Class D requires tower communication
- LAANC authorizes for drones
- Approval not guaranteed

Class B Airspace - Busy Airports

Class B airspace surrounds the busiest airports in the United States and is depicted with solid blue lines on sectional charts. It extends from the surface up to approximately 10,000 feet MSL. Major Class B airports include New York (JFK), Los Angeles (LAX), Chicago (ORD), and other high-traffic locations. Class B has multiple layers extending outward from the airport, with the innermost layer starting at the surface and outer layers starting at various altitudes. Entry into Class B requires explicit ATC clearance obtained by radio communication. For Part 107 drones without radios, LAANC authorization serves as clearance. LAANC availability around Class B airports is more reliable than other areas due to the critical nature of these locations. Class B airspace is highly structured, and operations must be precisely coordinated with ATC. Any deviation from cleared flight path is a serious violation. Operations in Class B require careful planning and strict adherence to the authorization.

KEY FACTS

- Surrounds busiest airports
- Surface to 10,000 feet MSL
- Solid blue lines on chart
- ATC clearance required
- Strict adherence to clearance

Class C Airspace - Medium Traffic Airports

Class C airspace surrounds medium-traffic airports and is depicted with solid magenta lines on sectional charts. The typical Class C floor is the surface with a ceiling around 4,000 feet AGL, though this varies by airport. Class C has two main components: the inner shelf (closer to airport, lower altitude limit) and outer shelf (farther, higher altitude limit). Examples of Class C airports include Denver, Phoenix, and many other significant but not top-tier airports. Entry into Class C requires two-way radio communication established with ATC before entry. For Part 107 drones, LAANC authorization is required instead of direct communication. The mode C veil (a 30nm radius where aircraft below 10,000 feet must have Mode C transponder equipment) extends around Class C, but Part 107 drones are exempt from this requirement. Class C airspace structure is less rigid than Class B, allowing more flexible operations with proper authorization.

KEY FACTS

- Medium-traffic airports
- Surface to ~4,000 feet AGL
- Magenta lines on chart
- Two-way radio communication required

- LAANC alternative for drones

Class D Airspace - Smaller Airports with Towers

Class D airspace surrounds smaller airports that have operational control towers and is depicted with dashed blue lines on sectional charts. Class D typically extends from the surface to 2,500 feet AGL within a 5nm radius of the airport. The tower frequency is shown on the chart (e.g., 119.5). Entry into Class D requires two-way radio communication with the tower established before entry, though clearance itself is not required (unlike Class B). For Part 107 drones, LAANC authorization may be available, but some smaller towers may not participate. If LAANC is not available, operators should contact the tower by phone to inform them of drone operations. Class D airspace is the most permissive of the controlled spaces, as contact is required but clearance is not. Many operations can proceed in Class D with proper notification and awareness of tower operations.

KEY FACTS

- Smaller airports with towers
- Surface to 2,500 feet AGL
- Dashed blue lines on chart
- Radio communication required
- Clearance not always required

Special Considerations for Class B, C, D

Operations in controlled airspace containing Class B, C, or D require careful planning. LAANC should be checked at least 24 hours before the intended operation, and authorization should be requested in advance when possible. Some areas have limited LAANC availability and may require alternative approaches. Direct communication with ATC may be possible for non-equipped aircraft to report position and intentions, enhancing safety awareness even in uncontrolled areas. Terminal Radar Service Areas (TRSA) may provide radar service in some areas even without Class C/D airspace. Weather can change rapidly in areas with active manned aircraft traffic, so continuous monitoring is important. High-traffic areas have many aircraft operations, and maintaining situational awareness and readiness to move is critical. If LAANC denies authorization, understanding why (conflict with scheduled flights, weather, etc.) helps in planning alternative approaches.

KEY FACTS

- Check LAANC 24 hours before
- Request auth in advance
- TRSA may provide radar service
- Weather can change rapidly
- Denial requires alternative planning

Chapter 17 Class E & G Airspace

Class E Airspace - General Controlled Airspace

Class E airspace is controlled airspace that doesn't fit into classes B, C, or D categories. It is depicted with dashed magenta lines on sectional charts. Class E can exist as either a surface area (floor at ground level, depicted with dashed magenta) or as an upper area (floor at 1,200 feet or 700 feet AGL depending on location). Class E surface areas surround some smaller airports with instrument approaches. Class E transition areas ensure controlled airspace exists above uncontrolled Class G airspace, typically starting at 700 feet or 1,200 feet AGL. En route Class E airspace covers areas between major airports where instrument flight rules are applied. Class E typically extends to 17,999 feet MSL (just below Class A). For Part 107 operations, Class E surface areas require LAANC authorization in controlled airspace. Class E upper areas (above 400 feet AGL) do not affect Part 107 operations since drones are limited to 400 feet.

KEY FACTS

- Controlled airspace, non-major airports
- Dashed magenta on chart
- Surface or 700/1200 feet floor
- Upper Class E above operations
- LAANC may be required

Class G Airspace - Uncontrolled

Class G airspace is uncontrolled airspace where no ATC services are provided. It is the default airspace classification where no other class exists. Class G is depicted in white on sectional charts (not explicitly marked). Class G typically exists below Class E airspace. In areas where Class E extends to the surface, there is no Class G. In other areas, Class G extends from the surface up to 1,200 feet (or 700 feet in some areas) where Class E begins. Class G has fewer restrictions than controlled airspace but Part 107 rules still apply. VFR (Visual Flight Rules) minimums for Class G are less restrictive than Class E: 1 statute mile visibility and clear of clouds (no specific altitude separation). For Part 107, the higher minimums (3SM visibility, 500 feet below clouds) still apply. LAANC may not be available in Class G, though some areas have airspace authorization through other means.

KEY FACTS

- Uncontrolled airspace
- Depicted in white on chart
- No ATC services
- Default airspace classification
- Part 107 rules still apply

Differences Between Controlled and Uncontrolled

The key difference between controlled and uncontrolled airspace is the availability of ATC separation services. In controlled airspace, ATC separates aircraft and provides instructions. In uncontrolled airspace, pilots are responsible for maintaining separation through see-and-avoid. Controlled airspace has stricter VFR weather minimums (1,000 feet above and below clouds, 2,000 feet horizontal, 3 statute miles visibility in Class E; lower in Class G). Uncontrolled airspace has lower minimums (Class G: 1 statute mile visibility, clear of clouds). Right-of-way rules apply in both, but in controlled airspace ATC provides separation to eliminate conflicts. For Part 107, controlled airspace operations are more complex due to LAANC requirements, but provide the benefit of ATC awareness of drone operations. Uncontrolled airspace operations are simpler but require heightened pilot vigilance for traffic avoidance.

KEY FACTS
- Controlled = ATC separation - Uncontrolled = see-and-avoid - Controlled = stricter minimums - Class E = ATC service available - Class G = pilot responsibility

Finding Your Airspace Class

To determine the airspace class at your intended operation location, consult a sectional chart for your area. Sectional charts are available online (free from various sources) and clearly show airspace boundaries and class designations using color coding (blue for B, magenta for C, dashed colors for D/E, white for G). The chart also shows the floor and ceiling of airspace areas. Many online tools (like LAANC apps or airspace planning websites) allow you to input your location and get airspace information automatically. If you're unsure about a location, contact your local Flight Service Station by phone for confirmation. Understanding your local airspace is essential for flight planning. Different seasons may have temporary changes (TFRs) that are shown on NOTAMs. Saving or printing sections of your sectional chart showing your common operating areas aids in quick reference during planning.

KEY FACTS
- Sectional charts show all airspace - Online tools available - Color coding indicates class - Contact Flight Service if unsure - Check for TFRs and NOTAMs

Special Use Airspace Overview

Special Use Airspace (SUA) is designated airspace where activities are restricted or where operations are subject to special limitations. SUA is shown on sectional charts with specific designations. Types of SUA include Restricted Areas, Prohibited Areas, Warning Areas, Military Operations Areas (MOAs), and Military Training Routes (MTRs). These areas are not all completely off-limits but require awareness and caution. Restricted Areas contain non-military activities that are hazardous to non-participating aircraft (like firing ranges or explosives testing). Prohibited Areas prohibit all flight except by special authorization (like around the White House, Camp David, or nuclear power plants). Warning Areas are off U.S. coasts and indicate hazardous activities but flight is not prohibited. Part 107 pilots should avoid all SUA unless specifically authorized. Checking for SUA at your flight location is part of the preflight planning process.

KEY FACTS

- SUA marked on sectional charts
- Restricted = non-military hazards
- Prohibited = no flight allowed
- Warning = off coasts
- Avoid unless authorized

Military Training Routes (MTRs)

Military Training Routes are designated airspace used by military aircraft for low-level navigation training. MTRs are shown on sectional charts with labels like "VR1001" or "IR301" (VR = visual route, IR = instrument route). MTRs at low altitudes (below 1,500 feet AGL) are the most relevant to Part 107 operations. Military aircraft train at high speeds and low altitudes, creating a significant collision hazard. Operations in MTRs are not prohibited but are highly risky during military training. If military training is active, MTRs are identified in NOTAMs. Part 107 pilots should check NOTAMs for MTR activity status and avoid flying in active routes. Some MTRs operate only during certain hours or certain days. Direct contact with the adjacent military facility may provide information on activity status. Flying in an MTR during military operations is extremely dangerous.

KEY FACTS

- Marked on sectional charts
- Low-level training routes
- VR = visual, IR = instrument
- Check NOTAMs for activity
- Avoid during active training

Military Operations Areas (MOAs)

Military Operations Areas (MOAs) are designated airspace where military activities are conducted. Unlike Restricted or Prohibited areas, flight through MOAs is not prohibited, but operations are conducted at the pilot's own risk. MOAs are marked on sectional charts with black outlines and the MOA name. Military aircraft conducting activities in MOAs may be at any altitude and engaging in maneuvers. If military operations are underway, the MOA is depicted on the sectional chart and listed in NOTAMs. Some MOAs have ATC coordination to separate military from civilian traffic. For Part 107 drones operating at 400 feet AGL, MOAs are not always a concern, but awareness is important. Checking NOTAMs for MOA activity and avoiding active MOAs is the safest approach. Flight Service can provide information about MOA status if planning a flight near an MOA.

KEY FACTS

- Flight not prohibited
- Operations at pilot's risk
- Marked on sectional charts
- Check NOTAMs for activity
- ATC coordination available

Temporary Flight Restrictions (TFRs)

Temporary Flight Restrictions (TFRs) are issued by the FAA to restrict or prohibit flight in specific areas for defined periods. TFRs are typically issued for national security, VIP movements, major events, disaster relief, or hazardous activities like forest fires. All Part 107 pilots must check for TFRs before every flight. The FAA publishes TFRs on its website (faa.gov/tfr) where they can be searched by location or date. LAANC apps typically display active TFRs. NOTAMs also include TFR information. TFR violations can result in serious penalties including certificate revocation. TFRs can be issued with short notice and may change frequently during active situations (e.g., forest fires). A TFR may restrict all flight or may allow flight at certain altitudes or times with restrictions. TFRs typically have a 10nm radius around the prohibited point but can vary significantly. If a TFR exists near your intended operation, contact Flight Service for exact details.

KEY FACTS

- Checked before every flight
- Check faa.gov/tfr website
- Violations = serious penalties
- Can change with short notice
- Typical radius: 10nm from point

Chapter 19 **Reading Sectional Charts**

Chart Overview and Purpose

Sectional aeronautical charts are detailed maps used by pilots to plan and navigate flights. They show topography, airspace, navigational aids, airports, obstacles, and other relevant aviation features. Sectional charts are updated regularly (typically every 6 months) to reflect changes. A standard sectional chart covers an area roughly 85 by 60 nautical miles. Multiple sectional charts are available covering the entire United States (about 37 charts for the continental US). Charts are available free online from FAA and various aviation websites. The chart title identifies the region (e.g., "Chicago Sectional"). Marginal information on the chart includes chart date, magnetic variation, scale, and other reference data. Understanding how to read a sectional chart is essential for flight planning as it shows everything a pilot needs to know about the airspace.

KEY FACTS

- Detailed aviation maps
- Updated every 6 months
- Covers ~85x60 nautical miles
- 37 charts for continental US
- Free online from FAA

Airspace Depiction on Charts

Sectional charts use color and line styles to show airspace classes. Class B is shown with solid blue lines. Class C is shown with solid magenta lines. Class D is shown with dashed blue lines. Class E is shown with dashed magenta lines (surface areas) or as labeled transition areas. Class G is the default and shown in white. Numbers printed on the chart indicate airspace floors and ceilings. For example, "3000" in blue indicates the floor of Class E airspace is 3,000 feet MSL. The altitude is always shown in hundreds of feet. Special Use Airspace (Restricted, Prohibited, MOAs, Warnings) are clearly marked with specific symbols and labels. Military Training Routes are shown as thin lines with labels like "VR1001". TFRs are not shown on sectional charts but must be checked separately from the FAA website.

KEY FACTS

- Blue = Class B,D
- Magenta = Class C,E
- White = Class G
- Numbers = altitude in hundreds of feet
- Special areas clearly marked

Airports and Runways

Airports on sectional charts are shown with symbols indicating the type and services available. Controlled airports have a tower symbol (circle with slash). Uncontrolled airports are shown as open circles. Military airports are shown with a specific symbol. The airport symbol size indicates the importance/size of the airport. Runway configuration is shown schematically. The runway label includes the runway heading (magnetic). For example, a runway labeled "27" points toward 270 degrees (west). Multiple runways are shown if they exist. Elevation is printed near the airport symbol in feet MSL. The chart shows available services (fuel type, lighting, etc.) in abbreviated form. Tower frequency is printed in blue for controlled airports. Flight Service and ATIS frequencies may also be shown. Runway length can be estimated from the chart symbol size relative to scale.

KEY FACTS

- Circle = uncontrolled airport
- Tower symbol = controlled
- Runway heading labeled
- Elevation in feet MSL
- Tower frequency in blue

Navigation Aids and Obstacles

Navigation aids like VORs (VHF Omnidirectional Ranges) and NDBs (Non-Directional Beacons) are shown on sectional charts with specific symbols. VORs display the magnetic variation and frequency. While Part 107 drones typically use GPS, understanding navigation aids helps with understanding the chart. Major obstacles like tall towers, antennas, and mountains are shown. Heights of towers and tall structures are labeled. Elevation contours show terrain. Cities and towns are marked to help with geographic orientation. Lakes, rivers, and coastlines are shown for reference. Power lines are shown in some areas (thin lines). These features help pilots understand the terrain and identify potential hazards. Sectional charts are comprehensive reference documents with enormous amounts of information. Learning to interpret all the symbols takes practice but is essential for safe flight planning.

KEY FACTS

- VORs with frequency and variation
- Obstacles with heights labeled
- Elevation contours show terrain
- Cities and towns marked
- Power lines indicated

Chapter 20 **LAANC & Airspace Authorization**

What is LAANC?

LAANC stands for Low Altitude Authorization and Notification Capability. It is a system created by the FAA to provide airspace authorization for drone operations in controlled airspace below 400 feet AGL. LAANC allows remote pilots to request flight authorization through mobile apps rather than requiring direct communication with air traffic control (ATC). The system is designed to streamline approvals while maintaining airspace safety. LAANC was initially a collaboration between the FAA, industry, and airports but is now integrated into the standard FAA authorization process. LAANC automatically checks the requested operation against scheduled manned aircraft operations, restricted areas, and other safety factors. Approval is typically granted within minutes if the operation is compatible with air traffic. LAANC operates 24/7 in most areas, allowing pilots to request authorization any time.

KEY FACTS

- FAA authorization system for drones
- Available in most controlled airspace
- Real-time approval system
- Mobile app access
- 24/7 availability

How LAANC Works

To request LAANC authorization, a Part 107 pilot uses a LAANC-approved app (like UAS Traffic Management apps provided by various vendors). The pilot enters their operation details including date, time, location, altitude, duration, aircraft type, and operation purpose. The app communicates with the FAA ATC facility controlling that airspace. The system checks the operation against scheduled arrivals/departures, restricted areas, and safety constraints. If the operation is compatible, approval is granted automatically. If the operation conflicts with manned aircraft or other constraints, the request is denied. A denied request typically shows the conflict reason (e.g., "departure traffic 1400-1430Z"), and the pilot can request alternative times. Approved authorizations show the time window (usually matching the requested time but may be restricted). The pilot must receive authorization before launching the drone.

KEY FACTS

- Enter operation details in app
- System checks compatibility
- Approval in minutes
- Denials show conflict reason
- Authorization has time window

LAANC Coverage and Limitations

LAANC is available around many airports and in controlled airspace areas throughout the United States. Coverage is expanding but not yet universal. Major metropolitan areas and busy airports typically have LAANC coverage. Smaller airports and rural areas may lack LAANC availability. Coverage gaps exist in some regions, requiring alternative authorization methods or operations in uncontrolled airspace. The FAA website shows LAANC availability by location. If LAANC is not available for your intended operation, alternative approaches include requesting a waiver, contacting ATC directly by phone, or moving the operation to uncontrolled airspace. LAANC currently covers all altitudes up to 400 feet AGL. Future versions may extend to higher altitudes. The system is designed specifically for Part 107 operations and is the primary method of authorization in controlled airspace.

KEY FACTS

- Available nationwide but not complete
- Major airports well covered
- Coverage varies by location
- Alternatives exist for gaps
- Phone contact to ATC possible

Pre-Flight Airspace Authorization Checklist

A thorough airspace authorization process includes multiple checks before launch. First, identify the airspace class at your intended operation location using a sectional chart. If in controlled airspace (B, C, D, or E), request LAANC authorization at least 24 hours before the intended flight, allowing time for requests that may be denied. Check the FAA TFR website (faa.gov/tfr) for active Temporary Flight Restrictions. Search the NOTAM database for any notices affecting your area. Verify that any received LAANC authorization is still valid and covers your intended time window. For operations near MOAs or other special airspace, check NOTAM status for activity. If operating near an airport, note the tower frequency for awareness. Brief all crew members on the airspace class and any restrictions. Keep a printed copy of LAANC authorization on site during the operation. Document all airspace checks as part of flight logs. This systematic approach ensures no authorization requirements are missed.

KEY FACTS

- Check airspace class first
- Request LAANC 24+ hours prior
- Check TFR database
- Review NOTAMs
- Keep authorization on site